Enhancing Urban Flood Mitigation Using Predictive WebGL Digital Twins: A Case Study of Nairobi County (Capstone)

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Declaration and Approval

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of my knowledge and belief, the research proposal contains no material previously published or written by another person except where due reference is made in the research proposal itself.

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Abstract

Urban flash flooding in Nairobi poses severe threats to public safety, local businesses, and critical infrastructure. Despite the severity of these hazards, current emergency monitoring networks are fundamentally constrained. Municipal authorities generally rely on flat, static 2D paper maps and rigid tables provided by remote state agencies. These traditional tracking systems fail because they cannot show how floodwaters move, accumulate, and progress across different neighborhoods over time. Consequently, local emergency responders and community leaders lack access to immediate, actionable information, forcing them into a purely reactive and dangerous posture during extreme storm events.

To bridge this critical communication gap, this project designs and develops a live, interactive 3D virtual replica of the city, known as an Urban Digital Twin, connected to a high-performance web dashboard. To ensure efficient performance on standard office computers without long delays, the system replaces slow, computationally intensive physics models with a lightweight, data-driven machine learning network. The engineering pipeline uses historical Sentinel-1 satellite radar data to identify historical water accumulation zones and trains an advanced artificial neural network to map local rainfall indicators against terrain elevation grids. The dashboard processes these feeds in real time, projecting dynamic flood depths, building footprints with volumetric data, and infrastructure exposure overlays onto an interactive browser interface. By translating complex scientific variables into intuitive, color-coded visualizations, this diagnostic platform demystifies hazard tracking. This framework equips Nairobi's public authorities with true situational awareness, enabling them to deploy emergency resources proactively and transition to data-driven climate adaptation planning.

Keywords: Flood Prediction, Nairobi, Urban Digital Twin, Real-Time Forecasting, Geospatial Dashboard, Disaster Management, Machine Learning Accessibility

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List of Abbreviations

ANN – Artificial Neural Networks

API – Application Programming Interface

AWS – Amazon Web Services

CHIRPS – Climate Hazards Group InfraRed Precipitation with Station Data

CNN – Convolutional Neural Networks

DEM – Digital Elevation Model

ERD – Entity-Relationship Diagram

GeoJSON – Geographic JavaScript Object Notation

GIS – Geographic Information Systems

GPU – Graphics Processing Unit

HTML – Hypertext Markup Language

LOD2 – Level of Detail 2

LSTM – Long Short-Term Memory

MAE – Mean Absolute Error

OOAD – Object-Oriented Analysis and Design

PHP – Hypertext Preprocessor

PostGIS – Post Geographic Information Systems

RMSE – Root Mean Square Error

RNN – Recurrent Neural Networks

SAR – Synthetic Aperture Radar

UDT – Urban Digital Twin

USGS – United States Geological Survey

WebGL – Web Graphics Library

WFP – World Food Program

Chapter 1: Introduction

1.1 Background Information

According to the foundational climate models analyzed by (Berkhahn & Neuweiler, 2024), urban pluvial and fluvial flash flooding represents an escalating socio-economic and environmental hazard globally, creating a public safety crisis that is intensely magnified by the compounding pressures of rapid climate change and uncontrolled urban growth. As documented by Singha et al. (2024), implementing robust flood hazard modeling is crucial for supporting disaster resilience and long-term climate adaptation workflows globally. Berkhahn & Neuweiler (2024) note that in expanding sub-Saharan African cities such as Nairobi, this structural vulnerability is particularly acute due to dense informal settlements, high surface imperviousness from building materials, and aging municipal drainage networks. Historically, environmental monitoring systems have relied heavily on physically based hydrodynamic 1D-2D coupled models to map surface water trends. However, as argued by Berkhahn & Neuweiler (2024), while these classical frameworks achieve strong spatial accuracy, their reliance on solving dense, complex systems of shallow-water equations introduces a severe computational bottleneck. The resulting operational latency of these physical models renders them practically obsolete for real-time decision-making during flash flood events. During these sudden downpours, localized storm life cycles are incredibly short, and lead times are minimal, meaning emergency teams require immediate, reactive situational awareness that legacy computers simply cannot compute in time.

To bypass these computational barriers, modern researchers are highlighting a structural shift in hydrologic computing toward data-driven surrogate modeling, which utilizes artificial intelligence to mimic computationally expensive physical simulations. According to Berkhahn & Neuweiler (2024), by leveraging lightweight machine learning architectures, such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs), software programs can approximate the complex outputs of intensive hydrodynamic models in fractions of a second. Within these frameworks, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have emerged as premier tools because they excel at capturing complex, non-linear temporal dependencies, enabling them to track how water volumes accumulate and change sequentially over time during rainfall-runoff processes. Concurrently, according to Singha et al. (2024), the availability of multi-modal Earth observation data from satellites enables precise mapping of historical flood boundaries across vulnerable terrain. (Berkhahn & Neuweiler, 2024) specify that combining cloud-penetrating

Sentinel-1 Synthetic Aperture Radar (SAR) imagery with high-resolution Digital Elevation Models (DEMs) provides precise landscape parameters that govern where water naturally pools and flows across a county's catchments.

However, computing these advanced predictive analytics is only half the battle; translating this raw data into an immediate emergency response faces a persistent communication gap when shared with the public. According to the field observations compiled in *(PDF) A Web-Based Geovisualization Dashboard for Real-Time Flood Risk Assessment in Hulu Langat, Malaysia, 2025*, traditional flood warnings are usually distributed as static telemetry text or non-interactive charts that public authorities and ordinary residents struggle to interpret intuitively during a crisis. Environmental visual analytics addresses this operational disconnect by turning raw data into interactive pictures. As outlined by Chen et al. (2024), the Integration of real-time predictive machine-learning data with Web Geographic Information Systems (Web GIS) and browser-based Urban Digital Twins (UDT), which serve as 3D virtual replicas of the physical city, fosters heightened community awareness and shared risk literacy. According to Sajja et al. (2026), embedding real-time sensor streams alongside automated, hardware-accelerated WebGL shader constructs a live, three-dimensional digital counterpart of the built environment directly in a web browser. This collaborative visualization framework shifts public safety operations away from blind, reactive disaster responses into an evidence-based, proactive model of risk reduction, dramatically strengthening urban community resilience.

1.2 Problem Statement

Urban flash flooding in Nairobi poses a major threat to public safety, critical infrastructure, and local economic activities. However, existing flood management systems are limited in their ability to support real-time decision-making. Traditional physics-based flood models are computationally expensive and too slow for emergency response, while data-driven surrogate models only generate static flood extent maps that fail to show the temporal progression, direction, and movement of floodwaters (Berkhahn & Neuweiler, 2024). As a result, emergency responders lack timely and actionable information needed to anticipate flood impacts and coordinate effective interventions.

Furthermore, current flood communication platforms often present hydrological data through spreadsheets or static 2D maps, making it difficult for emergency responders, local authorities, and communities to understand real-time flood risks. These systems lack building-level visualization, scenario simulation capabilities, and interactive tools for identifying vulnerable infrastructure (Chen et al., 2024). Consequently, disaster response remains largely reactive rather than proactive. Therefore, there is a need for an integrated system that combines real-time predictive deep learning models with an interactive 3D geospatial dashboard to improve flood visualization, situational awareness, and emergency preparedness in Nairobi (Sajja et al., 2026).

1.3 Research Objectives

1.3.1 General Objective

To develop an LSTM-powered 3D Urban Digital Twin dashboard for real-time flood forecasting and risk assessment in Nairobi County.

1.3.2 Specific Objectives

i. To analyze the current state of hydrologic cyberinfrastructure and data-driven surrogate modeling for urban flood prediction.

ii. To study and evaluate existing web-based hydrologic visualization systems and their application in flood monitoring and risk communication.

iii. To identify key gaps and limitations in current hydrologic forecasting and visualization systems.

iv. To develop and design an LSTM-based surrogate model and a hardware-accelerated 3D Urban Digital Twin (UDT) dashboard for near-real-time flood prediction and visualization.

v. To test and evaluate the predictive accuracy and computational performance of the proposed UDT system under extreme storm scenarios.

1.4 Research Questions

i. What is the current state of hydrologic cyberinfrastructure and data-driven surrogate modeling for urban flood prediction?

ii. How have existing web-based hydrologic visualization systems been applied in flood monitoring, and what are their strengths and limitations?

iii. What are the key gaps and challenges in current hydrologic forecasting and visualization systems?

iv. How can an LSTM-based surrogate model and a 3D Urban Digital Twin system be developed for near-real-time flood prediction and visualization?

v. How effective is the proposed UDT system in terms of predictive accuracy and computational performance under extreme storm conditions?

1.5 Justification

The practical and immediate value of this study lies in its ability to mitigate a major socioeconomic threat to Nairobi's residents. According to Berkhahn & Neuweiler (2024), standard flood forecasting strategies remain tightly bound to heavily funded, well-resourced institutions due to the massive computing power requirements and steep technical learning curves. By replacing these restrictive systems with a lightweight, browser-native surrogate model, this project democratizes access to high-fidelity flood intelligence. This software approach enables the platform to run efficiently on standard office hardware commonly found in local Kenyan administrative centers. With this accessible framework, municipal responders and neighborhood leaders can proactively deploy rescue resources. Instead of waiting for a disaster to unfold, they can execute targeted evacuations based on automated system alerts before peak initialization and pooling occur, significantly reducing the devastating financial and human impacts of urban flooding.

From an academic perspective, this project advances both environmental informatics and visual web analytics. According to Chen et al. (2024), this framework successfully demonstrates the interoperability of complex 2D and 3D datasets within a single, unified web-based layout. This visual integration enables users to easily see and interpret vertical vulnerabilities, such as structural data and flood heights, across a wide range of spatial and neighborhood scales. By bridging the traditional gap between raw mathematical data computation and human-centered design, the system creates a highly replicable, open-source template tailored for resource-constrained cities across the global South. Ultimately, this research provides the definitive engineering foundation needed to transition public safety infrastructure away from blind, reactive emergency responses toward a proactive, data-driven model for long-term climate adaptation planning.

1.6 Scope

This research operates within specific structural boundaries, geographically targeting high-exposure urban corridors and primary flood-prone catchments within Nairobi County. The predictive modeling relies on an LSTM surrogate trained with historical Sentinel-1 SAR (2019–2026), 30-meter elevation grids, and CHIRPS rainfall data to efficiently estimate surface runoff and water depths. For visualization, the system employs a Python-backed dashboard (Plotly Dash/Pydeck) featuring LOD2-level volumetric buildings, scenario sliders, and real-time telemetry from public meteorological APIs. Finally, the platform functions strictly as an informational tool, explicitly excluding sub-surface drainage hydraulics, groundwater infiltration, pipe erosion modeling, and automated emergency dispatch capabilities.

1.7 Limitations and Delimitations

1.7.1 Limitations

The primary operational limitation of this study is its total dependency on the uptime, data integrity, and structural continuity of external public open data APIs (such as Google Earth Engine, USGS, and local climate streams). Gaps, anomalies, or telemetry latency in these data feeds will directly propagate to the dashboard's visualization layer. Additionally, because the machine learning component serves as an AI surrogate network trained on pre-simulated scenarios, the system is fundamentally bound to approximation rather than to precise physical modeling; its predictive accuracy may decline when subjected to unprecedented, extreme anomalies that fall entirely outside the historical boundaries of the training dataset.

1.7.2 Delimitations

The research intentionally limits its focus to surface-level pluvial and fluvial flash flooding events. Alternative flood hazards, such as coastal storm surges or structural dam failures, are explicitly excluded because they do not align with the geomorphic context of the Nairobi region. The software stack is restricted to open-source, browser-native development environments (Python, WebGL, and JavaScript libraries) to guarantee that the final tool can be deployed and maintained by under-resourced public agencies without incurring proprietary licensing costs.

Chapter 2: Literature Review

2.1 Introduction

This chapter offers a critical synthesis of the scholarly literature, deep learning frameworks, and spatial architectures underpinning the Nairobi County Urban Digital Twin (UDT) flood-forecasting system. Aligned with the study's research objectives, the review begins in Section 2.2 by examining the paradigm shift from computationally intensive, physics-based hydrodynamic models to high-velocity AI surrogate networks. Section 2.3 then benchmarks existing web-based environmental dashboards, laying the foundation for Section 2.4 to identify critical, unaddressed gaps in real-time processing latency and reliance on infrastructure. Finally, Section 2.5 synthesizes these insights into a comprehensive Conceptual Framework that maps multimodal data ingestion, predictive LSTM modeling, and WebGL rendering pipelines engineered to overcome current limitations and support proactive urban climate adaptation.

2.2 Current State of Hydrologic Cyberinfrastructure and Surrogate Modeling

Hydrologic modeling has undergone a fundamental transformation driven by computational advances, shifting from traditional, physics-based numerical modeling to high-speed, data-driven surrogate frameworks (Berkhahn & Neuweiler, 2024). This transition marks a critical paradigm shift, particularly for urban emergency management, where the speed of data processing directly determines the effectiveness of disaster response (Berkhahn & Neuweiler, 2024). As computational limits have traditionally bottlenecked real-time forecasting, integrating advanced artificial intelligence architectures offers a viable pathway to overcoming these historical constraints (Sajja et al., 2026).

2.2.1 Overview of Hydrologic Cyberinfrastructure

Hydrologic surrogate models serve as computationally lightweight digital twins, virtual replicas of physical environments that learn nonlinear mappings from inputs, such as precipitation, slope, and land cover, to outputs, such as flood depth and volume. By recognizing these complex patterns, surrogates bypass computationally intensive calculations, enabling near-instantaneous forecasting (Berkhahn & Neuweiler, 2024; Sajja et al., 2026)).

To contextualize these advancements, it is necessary to evaluate the evolution of current modeling approaches. Traditionally, flood forecasting has relied on physically based numerical models that solve dense systems of differential equations to simulate water flow, precipitation, and terrain dynamics, grounded in empirical laws of fluid mechanics. To overcome the inherent processing delays of these numerical models, modern approaches leverage deep learning surrogates (Singha et al., 2024). These methods use advanced neural architectures, such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs), to enable high-speed inference (Singha et al., 2024). However, because these algorithms often function as isolated mathematical backends, the current engineering frontier lies in developing applied prognostic pipelines that translate these raw tensor outputs into interactive, real-time geovisualization platforms (Sajja et al., 2026).

The efficacy of these surrogate models is marked by both significant successes and persistent limitations. On the one hand, replacing heavy numerical computations with surrogate models drastically reduces processing latency. This efficiency effectively democratizes access to life-saving, real-time flood intelligence, allowing resource-constrained cities in the Global South to run complex simulations on standard office computers (Berkhahn & Neuweiler, 2024). On the other hand, data-driven surrogates face their own operational hurdles. First, these models rely heavily on large datasets of pre-simulated training scenarios to function accurately. Second, the raw, complex data they produce often remains disconnected from the interactive 3D dashboards that municipal authorities need to make rapid, informed decisions during a crisis (Chen et al., 2024).

2.2.2 Key Challenges in Existing Hydrologic Modeling Systems

Existing hydrologic modeling systems face several critical challenges that hinder their operational deployment, with computational latency a primary concern. Traditional physically based 1D-2D coupled models demand massive computational resources. This requirement creates a severe processing bottleneck, rendering them largely unsuitable for real-time decision-making during sudden urban flash floods, when lead times are minimal and rapid responses are essential (Berkhahn & Neuweiler, 2024).

Furthermore, these systems struggle to address the inherent spatiotemporal complexity. Accurately predicting flood dynamics requires models that capture complex temporal dependencies, such as tracking how water volumes accumulate over time, and robust spatial feature extraction to identify flow direction from high-resolution satellite radar and elevation grids(Singha et al., 2024). Failure to employ advanced deep learning architectures, such as RNNs, LSTMs, or CNNs, to manage these variables fundamentally limits a system's predictive integrity (Gude et al., 2020).

Compounding these modeling difficulties are severe hardware and memory constraints. Generating high-dimensional flood maps requires substantial resources. Without integrating model compression techniques, such as using autoencoders to reduce outputs to latent representations, these forecasting systems cannot operate efficiently on standard municipal hardware (Berkhahn & Neuweiler, 2024) These technical limitations perpetuate a digital divide in the Global South. Rapid urban growth, dense informal settlements, and aging infrastructure continually exacerbate flood vulnerability in these developing regions (Berkhahn & Neuweiler, 2024). However, these localities often lack the computing power of wealthy, industrialized regions required to run traditional, resource-intensive digital solutions, highlighting an urgent and unmet need for "frugal" yet highly robust digital twin alternatives (Berkhahn & Neuweiler, 2024; Sajja et al., 2026)).

2.3 Related Works in Hydrologic Visualization and Forecasting

This section examines recent developments in flood prediction and geospatial visualization systems.

2.3.1 Hydro3DJS: Web-Based 3D Visualization

Sajja et al. (2026) developed Hydro3DJS, a lightweight, browser-native 3D visualization library for rendering real-time watershed dynamics. By leveraging JavaScript, WebGL, and the Google Maps API, the framework integrates real-time telemetry from authoritative sources to support terrain-aware water animations and customizable 3D infrastructure models (Sajja et al., 2026). However, the system faces significant operational constraints. Hydro3DJS primarily functions as a visualization layer rather than an analytical engine, requiring tight coupling with external frameworks for computational tasks, and its rendering performance degrades when handling complex GeoJSON datasets (Sajja et al., 2026).

2.3.2 Hulu Langat Flood Risk Dashboard: Real-Time Risk Assessment

Similarly, to address the need for localized risk communication, the Hulu Langat Flood Risk Dashboard was developed as a two-tier Web-GIS platform to provide real-time risk assessments in Malaysia. It uses a PHP backend for data acquisition and a JavaScript frontend for geovisualization, and it introduces a dynamic multi-factor risk-scoring algorithm that evaluates variables such as river velocity and 24-hour rainfall intensity. Despite its utility, the Hulu Langat system remains strictly diagnostic rather than predictive, and its reliability depends entirely on the continuous uptime of the external JPS Selangor API. *(PDF) A Web-*

Based Geovisualization Dashboard for Real-Time Flood Risk Assessment in Hulu Langat, Malaysia, 2025.

2.3.3 Near-Real-Time Probabilistic Flood Forecasting Tool

To move beyond purely diagnostic monitoring, parallel research has focused on integrating predictive and probabilistic modeling into early warning systems. Mosimann et al. (2025) introduced a Near-Real-Time Probabilistic Flood Forecasting Tool for the Swiss context. Built on standard web technologies (HTML, JavaScript, CSS) and powered by a GeoServer interface connected to a PostgreSQL/PostGIS database, the tool visualizes potential inundation areas using precomputed probabilistic forecast members (Mosimann et al., 2025). This framework allows stakeholders to explore scenarios, including flood depths, hazard classes, and potential damage estimates, providing a proof-of-concept for navigating forecast uncertainties through interactive 2D maps and charts (Mosimann et al., 2025). Despite these conceptual advances, critical gaps persist. The Swiss forecasting tool remains restricted to selected stakeholders due to its prototype status, and its machine learning methods lack broad transferability across major river networks (Mosimann et al., 2025). Furthermore, these surrogate frameworks struggle to reconcile highly divergent stakeholder needs, as the data requirements for insurance companies differ significantly from the immediate, high-resolution geospatial intelligence required by local emergency responders (Mosimann et al., 2025).

2.4 Gaps in Related Works

A critical synthesis of the reviewed environmental systems reveals persistent technical limitations in hydrologic forecasting and geovisualization. To ensure practical viability and operational success, this study explicitly targets and resolves three primary gaps in the existing literature.

First, current web-based systems lack real-time, predictive forecasting. Most data-driven approaches, such as the Hulu Langat Flood Risk Dashboard, function purely as diagnostic monitors of current conditions based on existing API feeds *(PDF) A Web-Based Geovisualization Dashboard for Real-Time Flood Risk Assessment in Hulu Langat, Malaysia, 2025.*Even when surrogate flood models achieve accelerated computing speeds, they are often limited to generating static maps of maximum-extent inundation or rely entirely on precomputed scenario databases (Berkhahn & Neuweiler, 2024; Mosimann et al., 2025). This leaves a critical void in near-real-time spatiotemporal forecasting during live pluvial events. The present study addresses this constraint by configuring a Long Short-Term Memory

(LSTM) deep learning surrogate model designed to predict dynamic, near-real-time temporal variations in water depth and runoff volumes.

Second, a severe spatio-temporal disconnect persists in current environmental visualization tools. Traditional web-GIS platforms rely primarily on flat 2D maps or coarse raster rendering, which inherently fail to capture the fluid, continuous progression of floodwaters over time (Berkhahn & Neuweiler, 2024). This loss of resolution obscures vital micro-topographic vulnerabilities and localized structural blockages, ultimately impeding emergency responders' ability to accurately assess street-level exposure. This project bridges the visual gap by designing a browser-native 3D Urban Digital Twin (UDT) using WebGL shaders and Plotly Dash, which renders volumetric Level-of-Detail 2 (LOD2) building footprints and dynamic, interactive hazard layers within a unified layout (Chen et al., 2024; Sajja et al., 2026).

Finally, state-of-the-art forecasting systems are heavily burdened by infrastructure and computing dependencies. High-performance visualization engines often rely on rigid, expensive cloud infrastructures or tightly coupled computational libraries, effectively excluding under-resourced municipalities from using them (Chen et al., 2024; Sajja et al., 2026). Furthermore, traditional physically based numerical models demand massive computational resources, creating severe processing bottlenecks during the rapid lifecycles of sudden urban storms (Berkhahn & Neuweiler, 2024). This research directly addresses this digital divide by architecting a lightweight, hardware-accelerated rendering pipeline and an AI surrogate network that operate efficiently on standard office computing hardware without introducing rendering latency.

Table 2.1 Comparison of Existing Flood Forecasting and Visualization Systems

System	**Key Strength**	**Forecasting Capability**	**Visualization Capability**	**Critical Gaps**
Hydro3DJS	Browser-native 3D visualization with real-time telemetry integration	None (Visualization only)	Interactive 3D terrain and watershed rendering	Lacks predictive analytics and experiences performance degradation with complex spatial datasets
Hulu Langat Flood Risk Dashboard	Dynamic multi-factor flood risk	Diagnostic only (Current conditions)	Interactive Web-GIS risk maps	No future flood prediction and heavy dependence on external API availability

	assessment and real-time monitoring			
Probabilistic Flood Forecasting Tool	Probabilistic scenario analysis and uncertainty visualization	Precomputed probabilistic forecasts	Interactive 2D maps and charts	Limited transferability, restricted accessibility, and not suitable for real-time urban flood forecasting

2.5 Conceptual Framework

The conceptual framework for this study, as illustrated in Figure 2.1, provides a structured blueprint that integrates multi-source data ingestion, a deep-learning prognostic surrogate layer, and multi-faceted output interfaces. By replacing computationally expensive physics-based modeling with highly responsive AI inference, the system is designed to operate with high temporal and spatial fidelity on standard computing hardware. The architecture flows systematically from raw data inputs through integration and AI processing, culminating in actionable, user-facing visualizations.

The pipeline begins with the continuous ingestion of four distinct multimodal environmental input streams. The system captures Rainfall Data (such as daily CHIRPS), Topographic Data via high-resolution Digital Elevation Models (DEM), historical and spatial baselines from SAR Imagery (specifically Sentinel-1), and live telemetry from Real-time Sensors. These diverse inputs are channeled directly into a centralized Data Integration & Preprocessing module. Within this module, disparate spatial and temporal datasets are standardized and aligned, and critical local catchment characteristics, such as slope, curvature, and topographic indices, are extracted to fully prepare the data for predictive modeling (Singha et al., 2024).

After preprocessing, the unified data is fed into the core processing layer: the AI Surrogate Model. To avoid the severe numerical processing latencies typical of traditional 1D-2D coupled hydrodynamic models, this layer uses a combined Convolutional Autoencoder and Long Short-Term Memory (LSTM) network architecture (Berkhahn & Neuweiler, 2024). The autoencoder compresses high-dimensional spatial datasets to capture global terrain context, while the LSTM processes sequential temporal data to map nonlinear dependencies throughout a storm's lifecycle (Berkhahn & Neuweiler, 2024). This neural network directly powers the Flood Prediction Engine, which rapidly generates near-real-time spatiotemporal forecasts of runoff volumes and water depths.

Finally, the Flood Prediction Engine's prognostic outputs are integrated into a single, unified Integrated Urban Digital Twin (UDT) Geospatial Dashboard, which serves as the comprehensive interface for the outputs. This platform uses browser-native WebGL rendering to simulate terrain-aware mesh deformations and volumetric fluid depths without specialized local hardware (Sajja et al., 2026). Within the same layout, local planners can dynamically toggle combinations of return-period intensities and timeline projections via interactive scenario-exploration sliders (Chen et al., 2024). By consolidating these advanced visualization and decision-support features into one tool, complex hydrologic predictions and automated alert logs are seamlessly translated into actionable risk communication for emergency responders and strategic municipal planners (Chen et al., 2024). *(PDF) A Web-Based Geovisualization Dashboard for Real-Time Flood Risk Assessment in Hulu Langat, Malaysia, 2025*.

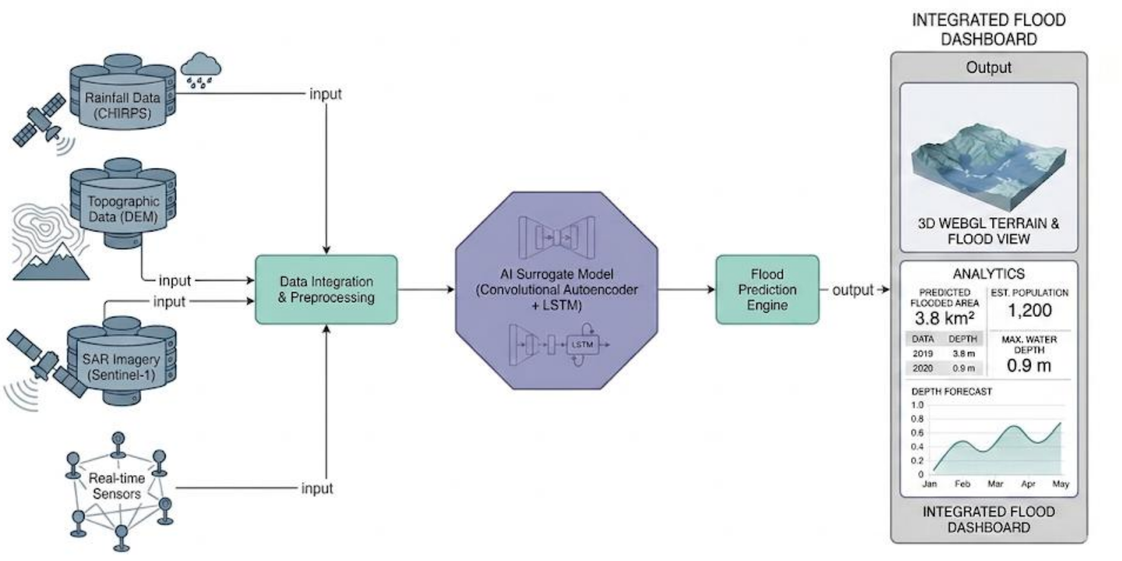

Figure 2.1 Conceptual Framework

Chapter 3: Methodology

3.1 Introduction

This chapter provides an overview of the methodological approach used in the study. It outlines the chapter's structure and briefly introduces key components covered in subsequent sections. To guide the architectural modeling, development, and integration of the WebGL geospatial dashboard, this project adopts the Object-Oriented Analysis and Design (OOAD) paradigm. OOAD supports an iterative, flexible, and modular development process, which is essential for effectively rendering complex 3D urban digital twins and handling dynamic, real-time data streams.

Furthermore, because the core predictive engine relies on an Artificial Intelligence (AI) component, specifically a Long Short-Term Memory (LSTM) network for flood forecasting, this section also introduces the experimental design used in the AI-based study. Unlike traditional software engineering, AI projects emphasize data-driven experimentation, model training, and validation. Therefore, this chapter covers critical stages, including data acquisition, data processing, model training, and testing. This ensures a systematic and rigorous approach to developing a highly responsive predictive solution for urban flood mitigation.

3.2 Research Paradigm

This project adopts an experimental research paradigm in which hypotheses regarding urban flood predictability and system rendering are validated through controlled data analysis and surrogate model training. The core methodology focuses on defining measurable forecasting objectives, collecting and preprocessing structured multimodal geospatial data, and systematically training and evaluating a deep learning surrogate model. For the Nairobi County flood mitigation system, this experimental approach is essential to verify whether an LSTM network can accurately approximate the results of complex physical hydrodynamic simulations under real-time constraints without exhausting local computing resources (Berkhahn & Neuweiler, 2024)

3.2.1 Data Acquisition

The project employs a structured data acquisition pipeline to collect and preprocess the multimodal datasets required to train and evaluate the AI-based flood prediction surrogate model. Sentinel-1 Synthetic Aperture Radar (SAR) imagery (2019–2026) is sourced from the Copernicus Data Space Ecosystem (*Copernicus Data Space Ecosystem | Europe's Eyes on Earth*, 2023), while the 30-meter-resolution Digital Elevation Model (DEM) is obtained via the Open Topography portal (*OpenTopography - Copernicus GLO-30 Digital Elevation Model*, 2021) through the Digital Earth Africa platform (*Digital Earth Africa | Unlocking the Power of Satellite Data*, 2019). Daily rainfall estimates are drawn from the CHIRPS dataset, accessed through the WFP Humanitarian Data Exchange (*Kenya: Rainfall Indicators at Subnational Level | Humanitarian Dataset | HDX*, 2023). Together, these datasets capture the rainfall dynamics, terrain morphology, and surface water behavior necessary for modeling urban flash flooding in Nairobi County. All acquired data undergo preprocessing, including spatial clipping, normalization, and quality filtering, to meet the standards required for high-fidelity geospatial and hydrological modeling.

3.2.2 Data Processing

Once the raw environmental data is acquired, it undergoes a comprehensive preprocessing pipeline to transform it into a standardized format suitable for the AI processing engine. This stage begins with data cleaning, addressing missing values, handling anomalies, and filtering statistical outliers to maintain dataset integrity. The disparate spatial and temporal datasets, originating from different sources, are then standardized and aligned to a uniform coordinate system. Furthermore, critical localized catchment characteristics, including slope, surface curvature, and topographic indices, are extracted to augment the input features. Numerical variables, such as precipitation and elevation, are normalized using techniques like Min-Max scaling to ensure stability during neural network training. This systematic preprocessing ensures that the input fed into the LSTM and autoencoder architectures is consistent, thereby mitigating overfitting and facilitating effective feature extraction.

3.2.3 Model Training

The model training phase focuses on configuring and training a deep learning surrogate model to predict near-real-time temporal variations in water depth and runoff volumes. The system uses a combined Convolutional Autoencoder and Long Short-Term Memory (LSTM) network architecture to replace computationally intensive hydrodynamic simulations. The autoencoder component is trained first to compress high-dimensional spatial datasets into latent representations that capture global terrain context. Subsequently, the LSTM network processes sequential temporal data to model the complex, nonlinear dependencies of rainfall-runoff processes throughout the storm’s lifecycle. During training, key parameters such as learning rates, batch sizes, and the number of training epochs are optimized to maximize prediction accuracy and computational efficiency. Stratified sampling and cross-validation are used to continuously evaluate performance, while dropout layers are used to maintain robustness and prevent overfitting to the training data.

3.2.4 Model Validation and Testing

The trained surrogate model undergoes rigorous validation and testing to benchmark its predictive accuracy and computational latency under extreme return-period scenarios. An independent test dataset, deliberately excluded from training and hyperparameter tuning, serves as the final evaluation suite to objectively assess the model's performance on previously unseen geospatial and temporal data. Prediction accuracy is measured using standard statistical metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R-squared for continuous water depth estimation. In addition to quantitative accuracy, the system is tested for computational performance to ensure that the integrated machine learning model and dashboard rendering pipeline operate within real-time constraints on standard office computing hardware. This validation phase confirms the system's ability to provide proactive situational awareness and to deliver actionable flood intelligence during sudden urban storm events.

3.3 System Development Methodology

This project uses an Agile methodology, specifically the Scrum framework, to guide the development of the AI-integrated Urban Digital Twin and its geospatial dashboard. The development workflow is organized into repeating, time-boxed cycles called sprints, each targeting specific technical milestones such as data ingestion, deep learning model calibration, system integration, and frontend deployment. Although the work is performed as an individual effort, established Scrum practices, including structured sprint planning, systematic progress monitoring, regular sprint reviews, and retrospective analysis, are applied to ensure a disciplined process, manage technical complexity, and foster continuous improvement throughout the system's lifecycle. This framework is essential for building the dashboard and the underlying LSTM predictive engine in manageable increments, ensuring that the integration of sophisticated machine learning components into the web interface remains stable and highly functional.

3.3.1 Justification of Methodology

Adopting an Agile development approach using the Scrum framework is well suited to the demands of this research, which requires continuous testing, iterative model refinement, and ongoing performance validation. Agile practices are particularly effective for Artificial Intelligence projects, providing the flexibility to make rapid adjustments during critical phases such as model training and performance tuning. According to Hutter et al. (2015), the iterative nature of hyperparameter tuning and model configuration benefits significantly from a structured yet adaptable development process. Furthermore, Berkhahn & Neuweiler (2024) highlight that data-driven approaches require frequent validation and retraining cycles to ensure the model remains accurate as environmental data patterns change. Since this project involves training a complex Long Short-Term Memory (LSTM) network and deploying it within a hardware-accelerated, web-based 3D visualization environment, the Scrum framework provides a transparent structure for managing tasks and responding to performance benchmarks or integration challenges as they emerge throughout the development process.

3.4 Methodology Diagram

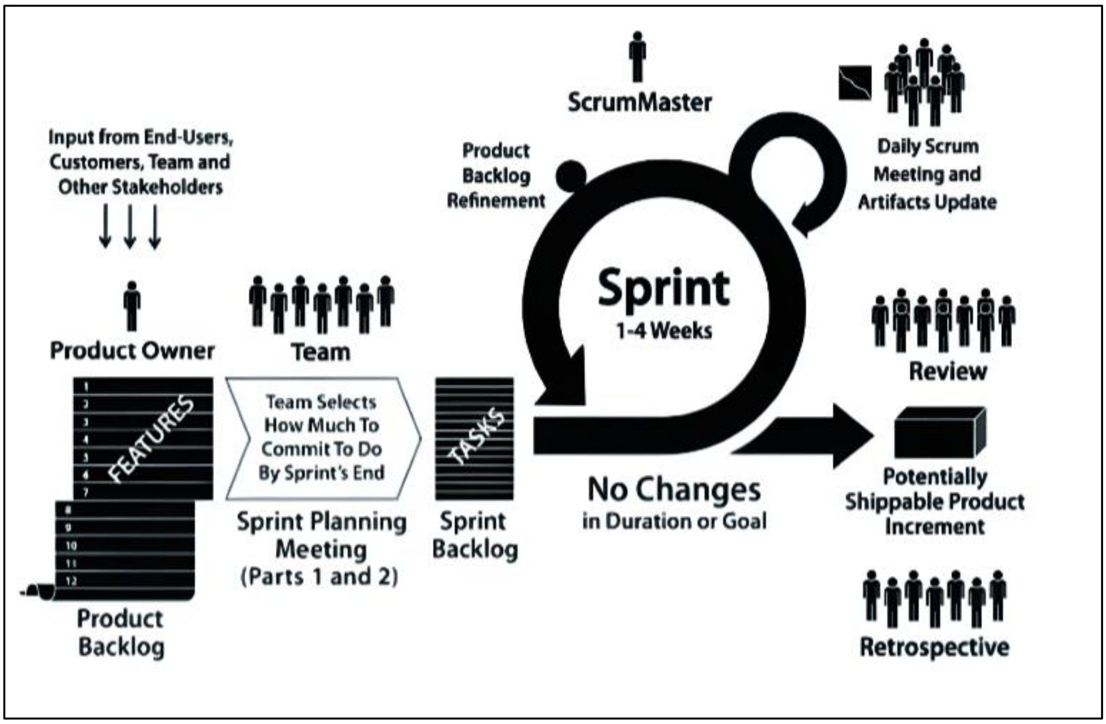

Figure 3.1 Scrum Framework (Kaur et al., 2021)

3.4.1 Product Backlog

This serves as the master repository for all anticipated system capabilities, technical requirements, and overarching project milestones. For the development of the Urban Digital Twin (UDT), this exhaustive ledger contained objectives ranging from Sentinel-1 SAR data ingestion and LSTM neural network configuration to the deployment of the interactive geospatial dashboard.

3.4.2 Sprint Planning Meeting

Prior to launching a new development cycle, specific items are strategically extracted from the main repository to form the focus of the upcoming phase. During these planning sessions, distinct technical targets are established and scoped, such as calibrating the predictive model's hyperparameters or designing the frontend map interface.

3.4.3 Sprint Backlog

This represents the dedicated inventory of tasks committed to completion within a single iteration. It breaks down the broader planning goals into granular, actionable engineering steps required to achieve the core objective of that specific cycle.

3.4.4 Sprint

This is the active, time-boxed window dedicated solely to engineering and implementation. Throughout this fixed-duration phase, the established objectives are locked to prevent scope creep, ensuring uninterrupted focus on delivering the committed elements of the flood assessment platform.

3.4.5 Daily Scrum Meeting

This practice involves brief, recurring synchronization to evaluate workflow velocity and pinpoint potential technical roadblocks. As an independent researcher, I maintained daily development logs and conducted structured self-assessments to ensure alignment with the overall project timeline.

3.4.6 Review

Upon concluding the iteration, the newly developed components undergo rigorous evaluation against the initial specifications. In this context, this includes tasks such as validating the LSTM model's predictive precision against historical data or verifying the rendering efficiency of the dashboard's spatial layers.

3.4.7 Retrospective

Following the review, a critical analysis of the development methodology is conducted to identify procedural inefficiencies. This reflective practice ensures continuous optimization of technical execution and time management for subsequent project phases.

3.4.8 Potentially Shippable Product Increment

Each completed iteration yields a functional, validated asset that integrates seamlessly into the broader architecture. These accumulated deliverables, whether a finely tuned surrogate model or an interactive mapping component, progressively build the final digital twin solution.

3.5 Analysis Diagrams

Analysis diagrams serve as visual blueprints during the requirements analysis phase, translating user needs into concrete system behaviors. These models bridge the conceptual gap between municipal stakeholders' expectations and the Urban Digital Twin's technical design, ensuring a clear understanding of the platform's intended functionality.

3.5.1 Use Case Diagram

This diagram identifies the platform's primary operators and their interactions. For this flood prediction project, the key actors include municipal planners, emergency responders, and system administrators. The diagram outlines major activities, including manipulating weather scenario sliders, visualizing 3D flood projections, and retrieving localized hazard summaries. This visualization is crucial for defining the operational boundaries of the geospatial dashboard and ensuring that all critical public safety workflows are supported.

3.5.2 Sequence Diagram

The sequence diagram maps the chronological exchange of information among the user, the dashboard interface, and the backend prediction engine. For example, it shows the exact sequence triggered when a user inputs a specific rainfall parameter: the frontend sends a request, the LSTM surrogate model processes the relevant spatial and temporal data, and the system returns a rendered flood-depth overlay. This helps visualize the logical flow of operations required to achieve real-time simulation without rendering delays.

3.5.3 Entity-Relationship Diagram (ERD)

The ERD represents the underlying database structure required to manage the platform's diverse datasets. It defines entities such as User Profiles, Meteorological Sensor Feeds, Topographic Layers, and Simulation Scenarios, and details how these records relate to one another. This blueprint guides the systematic organization, storage, and retrieval of multimodal data during high-speed forecasting operations.

3.5.4 Class Diagram

Using the Object-Oriented Analysis and Design (OOAD) approach, the class diagram details the software architecture by defining objects, attributes, and methods. It identifies key computational classes, including GeospatialRenderer, SurrogateModel, and TelemetryData. This structural guide supports the programming phase by mapping out how backend modules and frontend interfaces will be built and connected within the codebase.

3.5.5 Activity Diagram

This diagram outlines the step-by-step execution of core system processes. It maps the logical workflow of an emergency responder logging in to the portal, selecting a specific Nairobi catchment, initiating a flood prediction simulation, and viewing the resulting 3D risk assessment. This user-centric mapping clarifies the decision logic from start to finish, ensuring seamless, intuitive navigation.

3.6 Design Diagrams

Design diagrams advance the system's conceptual analysis into a concrete architectural framework. They refine earlier requirements by providing comprehensive schematics of the system’s layout, data structures, and technological interactions, ultimately guiding the final coding and deployment phases.

3.6.1 Database Schema

This section defines the precise storage architecture for the system's data. Given the reliance on geographic coordinates and real-time telemetry, the schema outlines how structured data, user settings, and geospatial layers (using spatial database frameworks such as PostGIS) are stored. The schema is designed to enable highly efficient spatial data retrieval, flexible scaling, and the real-time updates required for browser-native 3D mapping.

3.6.2 Wireframes

Wireframes provide a static visual representation of the dashboard’s user interface. These mockups map out the layout of essential screens, including the main 3D WebGL map viewport, the interactive control panel with environmental sliders, and the analytical summary overlays. Creating these visual blueprints ensures the interface remains intuitive and accessible for civil authorities before frontend development begins.

3.6.3 System Architecture

The system architecture diagram illustrates the end-to-end technology pipeline, showing how the browser-native frontend seamlessly connects to the computational backend. It maps the integration of the Plotly Dash and WebGL visualization layers, the secure API gateways for real-time sensor streams, and the LSTM neural network module that generates rapid flood predictions. This comprehensive overview confirms that the infrastructure is scalable, computationally efficient, and capable of processing high-speed environmental data.

3.7 Deliverables

This section details the primary outputs and operational artifacts generated throughout the lifecycle of the Urban Digital Twin (UDT) project.

3.7.1 Trained Predictive LSTM Surrogate Model

A foundational deliverable of this study is a fine-tuned recurrent deep learning surrogate engine optimized for rapid spatiotemporal flood forecasting. This long short-term memory (LSTM) neural network bypasses the high computational latency of traditional hydrodynamic frameworks by directly predicting multi-step-ahead water-level maps from live precipitation inputs. Trained, validated, and iteratively refined using simulated catchments and Sentinel-1 SAR observations, the final model offers top-tier generalizability and highly accelerated execution speeds to support tactical emergency management decisions.

3.7.2 Web-Based Geospatial Dashboard Application

The interactive dashboard serves as the consumer-facing interface, contextualizing raw environmental data and neural network outputs for end users. Built on a lightweight, browser-native platform, the implementation features hardware-accelerated 3D viewports that render volumetric building layers over animated, terrain-aware floodwater layers. It embeds reactive configuration widgets, such as event-scenario sliders and customized visualization popups, enabling civil defense authorities to interactively evaluate cascading infrastructure vulnerabilities without specialized computing hardware.

3.7.3 Technical System Documentation

A comprehensive documentation suite compiles all architectural specifications, data engineering workflows, and system design diagrams for the platform. This technical reference manual maps out the end-to-end data pipeline: from the initial ingestion of telemetry streams across standard data boundaries to client-side WebGL canvas adjustments. It includes explicit mapping guides for the spatial database schemas, structured continuous integration configurations, configuration guides for the underlying microservices, and deployment parameters to facilitate long-term maintenance and structural replication.

3.8 Tools and Techniques

The technical layout utilizes a curated software ecosystem chosen to manage the massive datasets and high-concurrency demands inherent in real-time geospatial modeling.

3.8.1 Google Colab & Cloud Infrastructure

For training and optimizing the deep learning surrogate architectures, Google Colab was selected as the baseline sandboxed computing environment. Its on-demand access to high-performance Graphics Processing Units (GPUs) significantly shortens compilation windows when manipulating dense Sentinel-1 matrices and time-series radar vectors, saving valuable time and removing dependencies on specialized on-premise components.

3.8.2 Python Programming Language

Python acts as the core technological glue across both the machine learning pipeline and full-stack web integration. This object-oriented ecosystem was critical to developing the data acquisition microservices, processing incoming GeoJSON vectors, configuring neural network architectures, and orchestrating reactive backend callback loops within the dashboard framework.

3.8.3 TensorFlow Deep Learning Library

TensorFlow, combined with its high-level Keras APIs, provides the computational engine for building the predictive surrogate model. It handles the tensor math and matrix updates required to train deep recurrent layers, enabling the network to capture complex, nonlinear dependencies between cumulative rainfall timelines and sequential surface water pooling patterns.

3.8.4 Plotly Dash & Pydeck Visual Frameworks

Plotly Dash was deployed as the declarative Python engine for managing the full-stack layout, structuring web components into an interactive grid with responsive Bootstrap templates. Pydeck provides high-performance bindings that interface with Deck.gl rendering pipelines, enabling the system to stream multi-layer geospatial entities smoothly within a standard viewport using parallelized WebGL processing.

3.8.5 PostgreSQL with PostGIS Extensions

To accommodate the spatial requirements of multi-hazard analysis, a PostgreSQL relational database with a PostGIS extension was implemented. This configuration manages the geographic properties of physical networks, critical asset vectors, and topographic indicators, supporting rapid geometric queries and feature selection by the processing backend.

3.8.6 Amazon Web Services (AWS) Deployment Suite

Cloud hosting services are managed on Amazon Web Services to build a scalable and resilient infrastructure footprint. The system uses containerized configurations to ensure uniform code execution across testing environments, while serverless computing clusters handle computationally intensive machine learning inference cycles without requiring permanent, costly server allocations.

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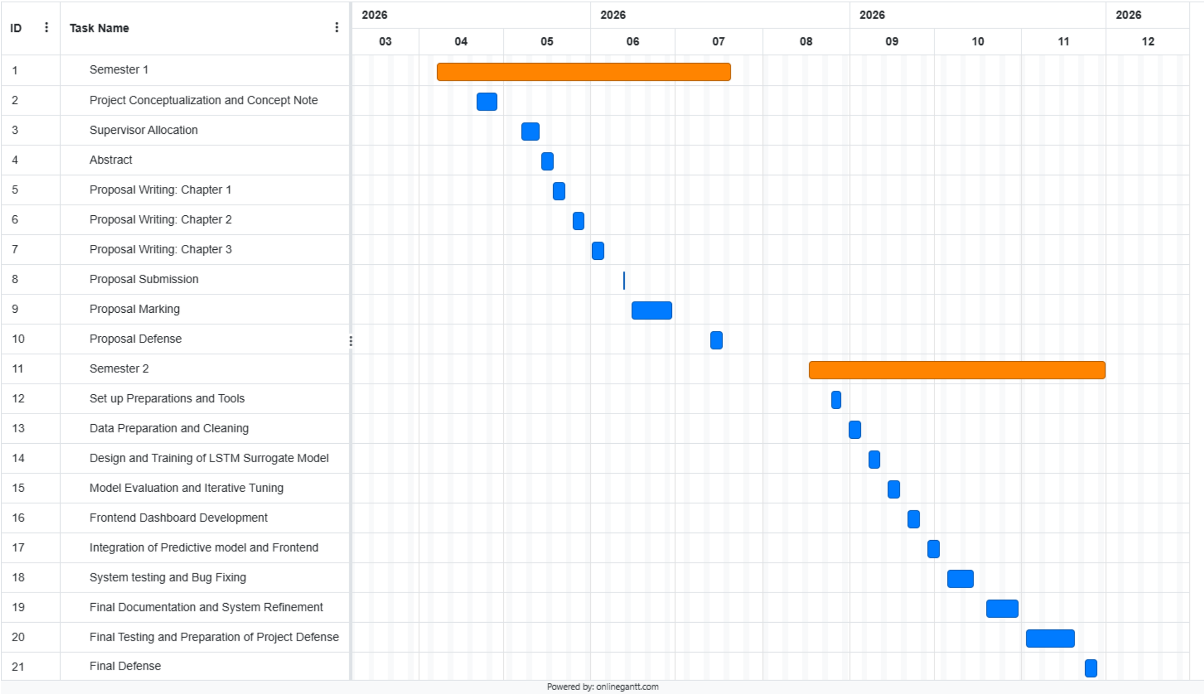
Appendix

Appendix 1: Gantt Chart



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